

Project Title	Clinical Trial Disease Category Classification Using NLP and Machine Learning
Skills take away From This Project	• Python Programming & Data Processing • CSV to DataFrame Conversion using Pandas • Medical Text Cleaning & Preprocessing • Natural Language Processing (NLP) Techniques • Tokenization, Lemmatization & Stop Word Removal • Exploratory Data Analysis (EDA) • Medical Text Classification using NLP • TF-IDF Vectorization & Feature Engineering • Machine Learning for Text Classification • Model Evaluation using Accuracy, Precision & Recall • Data Visualization using Matplotlib / Seaborn / Plotly • Clinical Trial Data Analysis • Streamlit Web Application Development • Machine Learning Model Deployment using Streamlit • End-to-End NLP Pipeline Development
Domain	Healthcare Analytics, Medical NLP & Clinical Trial Intelligence Systems

Problem Statement:

The healthcare industry generates large volumes of clinical trial data, including medical summaries, eligibility criteria, treatment details, and disease-related information. However, this textual data is often unstructured and difficult to analyze directly for extracting meaningful medical insights.

The challenge is to preprocess and transform the clinical trial text data into a structured format and apply Natural Language Processing (NLP) and Machine Learning techniques to classify clinical trial summaries into appropriate disease categories. Analyzing these categories helps in understanding disease-specific clinical studies and improves medical research organization.

Additionally, the project focuses on building an intelligent disease classification system using the “Brief Summary” column of clinical trial data to automatically predict disease categories based on medical text content. This system can support healthcare analytics, medical research, and AI-driven clinical trial management systems.

This project aims to build a complete **end-to-end data analytics and machine learning pipeline** that enables candidates to:

- Clean and preprocess mobile product data using Python
- Perform exploratory data analysis (EDA) to identify key trends and patterns
- Apply clustering techniques to segment products or users into meaningful groups
- Analyze each cluster to generate actionable business insights
- Build a similarity-based product recommendation system
- Develop an interactive application using Streamlit for visualization and recommendations

Objective:

This project aims to build a complete end-to-end NLP and machine learning pipeline that enables candidates to:

- Clean and preprocess clinical trial text data using Python
 - Perform exploratory data analysis (EDA) on medical datasets
 - Apply Natural Language Processing (NLP) techniques on clinical summaries
 - Classify clinical trial summaries into disease categories
 - Analyze disease-specific patterns and medical insights
 - Build a machine learning model for medical text classification
 - Evaluate model performance using classification metrics
 - Develop an interactive application using Streamlit for disease prediction and visualization
 - Deploy an end-to-end medical NLP classification system
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Approach:

1. Data Collection

- Import the clinical trial dataset in CSV format
 - Load the dataset into Python using Pandas DataFrame
 - Understand the dataset structure, columns, and data types
 - Select the “Brief Summary” column for NLP-based analysis
 - Identify disease category labels for classification
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2. Data Preprocessing

- Handle missing values and remove duplicate records
- Perform text cleaning and normalization
- Remove stop words, punctuation, and special characters
- Apply tokenization and lemmatization techniques
- Convert text data into numerical format using TF-IDF Vectorization
- Encode target labels for machine learning models

3. Exploratory Data Analysis (EDA)

- Analyze the distribution of disease categories
 - Identify the most common medical conditions and keywords
 - Explore relationships between clinical trial summaries and disease categories
 - Identify patterns, trends, and correlations in medical text data
 - Perform statistical summaries and text-based analysis
 - Visualize category distributions and word frequency patterns
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4. NLP-Based Disease Classification

- Apply Natural Language Processing (NLP) techniques
- Convert medical text into numerical features using TF-IDF Vectorization
- Train machine learning models for disease category prediction
- Classify clinical trial summaries into disease categories
- Compare multiple classification algorithms for performance analysis
- Visualize model performance using evaluation metrics and plots

5. Disease Prediction System

- Build a machine learning–based disease classification system
 - Use clinical trial “Brief Summary” text data
 - Apply NLP techniques such as TF-IDF Vectorization
 - Predict disease categories based on medical text input
 - Validate prediction accuracy and model performance
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6. Application Development & Visualization

- Build an interactive application using Streamlit
 - Create visualizations using Plotly / Seaborn / Matplotlib
 - Display disease category predictions and analysis results
 - Provide an interface for clinical trial summary prediction
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7. Insights & Reporting

- Generate insights on:
 - Disease category distribution
 - Frequently occurring medical terms
 - Clinical trial text patterns and trends
 - Model performance and prediction accuracy
 - Support healthcare analytics and medical research decision-making
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Dataset Link: [Clinical Trial Disease Dataset](#)

Results:

- Cleaned and preprocessed clinical trial dataset
- Successful CSV to DataFrame conversion using Pandas
- Well-structured medical text dataset for NLP analysis
- Effective disease category classification using machine learning
- Successful application of NLP techniques on clinical trial summaries
- Accurate prediction of disease categories from medical text
- Performance evaluation using Accuracy, Precision, Recall, and F1-Score
- Interactive Streamlit application for disease prediction and visualization
- Insightful data visualization using Plotly / Seaborn / Matplotlib
- End-to-end NLP and machine learning pipeline implementation

Project Evaluation metrics:

- Proper preprocessing of clinical trial text data
- Accurate CSV to DataFrame conversion using Pandas
- Effective Exploratory Data Analysis (EDA)
- Correct text cleaning, tokenization, and lemmatization
- Proper implementation of NLP techniques and TF-IDF Vectorization
- Accurate disease category classification using machine learning
- Performance evaluation using Accuracy, Precision, Recall, and F1-Score
- Effective visualization of medical data and prediction results
- Streamlit application functionality and user interface
- Clean, modular, and well-structured Python code
- Proper model integration and prediction workflow
- Insight generation and interpretation from clinical trial data

Technical Tags:

Python, Pandas, Data Preprocessing, Natural Language Processing (NLP), Text Classification, TF-IDF Vectorization, Tokenization, Lemmatization, Machine Learning, Clinical Trial Data Analysis, Exploratory Data Analysis (EDA), Feature Engineering, Data Visualization, Matplotlib, Seaborn, Plotly, Streamlit Application, Healthcare Analytics, Medical NLP, Disease Category Classification.

Project Deliverables:

- Clinical Trial Dataset (CSV Raw Data File)
- Cleaned and Preprocessed Dataset (CSV / DataFrame)
- Python Script for Text Preprocessing and NLP Operations
- TF-IDF Vectorization and Feature Engineering Implementation
- Machine Learning Model for Disease Category Classification
- Model Evaluation Report with Performance Metrics
- Data Visualization using Plotly / Seaborn / Matplotlib
- Interactive Streamlit Application for Disease Prediction
- EDA Report with Clinical Trial Insights and Patterns
- End-to-End NLP Pipeline Implementation
- Final Project Documentation File

Project Guidelines:

- Perform proper preprocessing of clinical trial text data
- Handle missing values and remove duplicate records
- Ensure accurate CSV to DataFrame conversion using Pandas
- Conduct meaningful Exploratory Data Analysis (EDA)
- Apply proper text cleaning, tokenization, and lemmatization techniques
- Use TF-IDF Vectorization for feature extraction
- Implement machine learning models correctly for disease classification
- Evaluate models using Accuracy, Precision, Recall, and F1-Score
- Use appropriate visualization tools (Plotly / Seaborn / Matplotlib)
- Develop a structured Streamlit application for disease prediction
- Maintain clean, readable, and modular Python code
- Generate meaningful insights from clinical trial data
- Provide proper project documentation and reporting

References:

Project Live Evaluation Guidelines	Project Live Evaluation
Project Excellence Series: Guided Learning & Problem Solving [Python & SQL](Tamil)	Project Excellence Series: Guided Learning & Problem Solving [Python & SQL](Tamil)
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	How to Use GitHub.pptx
Streamlit Reference English	Streamlit Recordings
Streamlit Reference Tamil	Streamlit Recording Tamil
Project Orientation (English)	
Project Orientation (Tamil)	Project Orientation Session Recordi...

Timeline:

The project must be completed and submitted **within 10 days from the assigned date.**

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Note: You can join the doubt session using the link provided in sheet given below

Timing: Monday-Saturday (3:30PM to 4:30PM)

Session Link : [Project Doubt Session \[DS/AIML\]](#)

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing: Monday-Saturday (05:30PM to 07:00PM)

Booking link : <https://forms.gle/1m2Gsro41fLtZurRA>

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